

Computational Signature Diagnostics for the AM–GM Boundary Program

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Abstract

The AM–GM Boundary program now has a scalar realization document and a finite-band typology document, but it lacks a dedicated computational leg. This memo proposes one. The role of the computational leg is not to provide visual demonstrations after the theory is finished. Its role is to make the signature machinery executable on controlled model classes; to discover where the formal hypotheses in the typology and weak-coupling arcs are sharp; to build an estimation bridge toward biological data; and to produce cross-diagnostics against neighboring boundary frameworks, especially Markov-blanket and Active Inference models. The central object remains the typed interface signature

$$\Sigma(S, P, \Delta, V) = (K, a, \varepsilon_{\text{scalar}}, \varepsilon_{\text{band}}, \mathcal{R}, C, \eta; J),$$

introduced in Document 2 Sections 2–3. The computational leg supplies two complementary pipelines: an *evaluation pipeline*, where the generator and readout are known and the signature is computed, and an *estimation pipeline*, where finite trajectories or partial observations are used to recover a signature with uncertainty. The recommended first deliverable is a notebook suite organized into four tracks: validation on known killed-process generators, perturbative boundary diagnostics under weak coupling, exploratory cross-stratum composition, and estimation/crosswalk projects aimed at Document 2’s Document 3 handoff. The output should be treated as program infrastructure and as a research bridge: it should feed back into Document 2’s open arcs F1–F4, sharpen Document 1’s perturbative Appendix H, and make the path from formal typology to empirical extraction less speculative.

Position in the sequence. This memo is a computational planning document for the mature AM–GM stack. Document 1 is the scalar realization: it constructs positive scalar interface values in the killed-process class, realizes the weighted geometric evaluator on spectrally compressed blocks, extends the synthesis to slow-fast Feynman–Kac interface values, and adds a perturbative weak-coupling appendix. Document 2 surrounds that scalar regime with a finite-band signature typology: rank-one scalarity, interpretable stratum 2a, formal-cone non-diagnostic cases, stratum 3 cone-valued comparison, and stratum 0. The present memo does not add a theorem. It describes a computational bridge between Document 1’s scalar model class, Document 2’s supplied-signature typology, and the future empirical extraction methodology of Document 3.

Reference convention. “Doc. 1” refers to *The AM–GM Boundary: A Spectral Realization of Geometric Composition Across Non-Transferable Blocks*, monograph v34. “Doc. 2” refers to *The Finite-Band Typology of Interface Signatures*, draft v16. Section references are internal section

references to those documents. “Memo 20” and “Memo 21” refer to the weak-coupling scalar-compression and Feynman–Kac product-defect memos now incorporated into Doc. 1 Appendix H.

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1 Diagnosis: why a computational leg is now needed

The program has crossed the threshold at which purely formal development is no longer enough to guide the next work. That is a good sign. Document 1 gives the scalar theory a concrete killed-process realization: outer geometric aggregation is characterized in Part II, interface-value obstructions are isolated in Part III, Gibbs-lifted killed processes supply the scalar realization in Part IV, and static plus slow-fast synthesis is established in Part V. Its Appendix H now gives a perturbative weak-coupling account, separating scalar compression integrity from composition integrity. Its Appendix I identifies the near-degenerate boundary where scalar compression degrades into finite-mode vector structure.

Document 2 then changes the program’s target object. The scalar interface value becomes the typed interface signature

$$\Sigma(S, P, \Delta, V) = (K, a, \varepsilon_{\text{scalar}}, \varepsilon_{\text{band}}, \mathcal{R}, C, \eta; J),$$

indexed by a system, partition, cycle length, and viability readout (Doc. 2 Sections 2.1–2.3). The typology assigns supplied signatures to regions rather than extracting them from raw biological data (Doc. 2 Sections 2.4 and 9.2). Its central technical work dissolves the candidate readout-insensitive

finite-band stratum 2b, leaving stratum 1, interpretable stratum 2a, the formal-cone non-diagnostic region, stratum 3, and stratum 0 (Doc. 2 Sections 3–4 and 10). The immediate problem is therefore no longer just whether the theory is conceptually coherent. The immediate problem is whether the signature machinery is operational.

The missing layer is computational empirics over model classes. Between formal proof and biological extraction there should be a controlled laboratory of synthetic killed-process, finite-band, cone-valued, weakly coupled, and partially observed systems. These systems should be simple enough that the ground truth is known, but rich enough to exercise the signature components. The computational leg should answer four questions:

1. Can the signature tuple be computed in model classes that instantiate the monograph and the typology?
2. Do the strata and defect quantities behave as expected when spectral gaps, cycle lengths, killing rates, barriers, readouts, coupling strengths, and tolerances vary?
3. Which of Document 2’s technical deferrals F1–F4 look theorem-shaped after simulation, and which look irreducibly compatibility-dependent?
4. Can finite-sample estimators recover the same signatures from trajectories, perturbations, and partial observations, thereby preparing the ground for Document 3?

The computational leg is therefore not a presentation layer. It is a hypothesis-discovery layer, a type-discipline audit layer, and a bridge to empirical methods.

2 What the computational leg is and is not

The computational leg should be framed as *Computational Signature Diagnostics for Killed-Process Individuality*. The internal shorthand “signature laboratory” is useful, but the public-facing phrase should emphasize diagnostics rather than program infrastructure. The goal is not merely to validate AM–GM-native claims for insiders. The goal is to produce reusable diagnostics for systems with absorbing failure, metastability, finite-band survival structure, and candidate individuality boundaries.

2.1 It is not a replacement for proofs

No notebook should be treated as evidence for Theorem II.1, Theorem II.2, the readout-rigidity theorem, or the A5-admissible classification theorem. Those are mathematical results. Computation cannot replace their proofs. What computation can do is expose whether the hypotheses behind those theorems are operationally checkable, numerically stable, and falsifiable on controlled classes.

2.2 It is not yet biological empirics

The first notebook suite should not start by modeling cells, colonies, or forests. That would add biological ambiguity before the signature machinery has been debugged. The first systems should be engineered finite-state killed processes: one-well chains, multi-well reversible chains, finite cone examples, coarse-state Feynman–Kac models, and weakly coupled product chains. These are not intended as biological claims. They are calibration devices.

2.3 It is a middle layer

The right stack is

$$\begin{aligned} \text{formal typology} &\longrightarrow \text{computational signature diagnostics} \\ &\longrightarrow \text{biological signature extraction.} \end{aligned}$$

The first arrow asks whether supplied signatures can be evaluated and classified in controlled models. The second arrow asks whether signatures can be estimated from data. Both are required before the program can responsibly make empirical claims about real systems.

3 Two pipelines: evaluation and estimation

The computational leg should split immediately into two pipelines. The distinction is load-bearing.

3.1 The evaluation pipeline

The evaluation pipeline assumes the mathematical model is known. Its abstract form is

$$\text{Eval} : (G, V, \Delta, \eta, \mathcal{R}, C) \longmapsto \Sigma,$$

where G is a finite killed generator or a family of such generators, V is the viability readout, Δ is the outer cycle, η is the declared tolerance, $\mathcal{R}$ is the admissible readout class, and C records composition-compatibility data. This pipeline is designed for Document 1 and Document 2 validation. It should compute eigenvalues, retained bands, scalarity defects, band defects, candidate cones, cone comparisons, basin-witness diagnostics, weak-coupling leakage, Feynman–Kac product defects, and typology assignments.

This is the pipeline behind the original proposed interface. It is essential, but it is not sufficient for Document 3.

3.2 The estimation pipeline

The estimation pipeline assumes the generator is not given. Its abstract form is

$$\text{Est} : (\text{trajectory samples, perturbation data, } \mathcal{R}, \eta, \text{observation model}) \longmapsto \hat{\Sigma} \text{ with uncertainty.}$$

This is the pipeline Document 3 needs. It must estimate low modes, retained dimension, metastable basins, defect terms, readout behavior, cone comparisons, and dynamic-regime labels from finite samples, partial observations, and perturbation responses. Doc. 2 Section 9.2 anticipates exactly this family of methods: sublethal perturbation tomography, basin reconstruction, hysteresis/multistability assays, and cone-comparability versus scalarization-dependent comparison tests.

The two pipelines should be developed together. Every validation notebook should eventually have an estimation companion: first compute the true signature from the known generator, then simulate trajectories, hide the generator, estimate the signature, and compare $\hat{\Sigma}$ to Σ . Without that benchmark loop, the computational leg will remain a toy-model exhibit rather than a Doc. 3 prototype.

4 Design principles for the notebook suite

4.1 Each notebook has a validation section and an exploration section

The validation section reproduces a theory-predicted behavior in a controlled system. The exploration section deliberately pushes beyond the theorem: boundary parameter regimes, noise, finite samples, weak coupling, partial observation, dynamic transitions, or mixed types. The validation section gives confidence that the machinery is implemented correctly. The exploration section is where new hypotheses should emerge.

4.2 Every notebook logs theory-computation disagreements

A surprising stratum assignment should not be dismissed as a numerical bug. It should be logged as one of four possibilities:

- (D1) implementation error;
- (D2) numerical conditioning or finite-sample artifact;
- (D3) misdeclared signature data, especially $\mathcal{R}$, C , or η ;
- (D4) genuine pressure on the formal theory.

The fourth case is the point of the lab. The formal-cone versus interpretable-2a boundary, dynamic stability of basin witnesses, and cross-stratum composition are especially likely to produce useful surprises.

4.3 All compatibility data must be explicit

The computational leg should treat C as first-class. If a scalar readout is supplied before pooling, say so. If a stratum-1 scalar is embedded into a cone, record the embedding. If basin coordinates are shared across blocks, record the coordinate convention. If a product-typed interface is retained, record the type. Doc. 2 Section 2.2 stresses that C is supplied compatibility structure, not a promise that the typology gives a canonical cross-block law. The notebooks should enforce that discipline.

4.4 Every scalar value must carry its type

The weighted geometric evaluator in Doc. 1 applies to positive scalar interface values once the scalar interface conditions are met. It does not automatically apply to finite-band vectors, basin-coordinate arrays, cone-valued states, or raw product survival probabilities. Appendix G of Doc. 1 and Example 7.2 of Doc. 2 make this distinction central: evaluator super-blocks and physical product super-blocks are different objects. The computational suite should preserve that type distinction in its plots, tables, and diagnostics.

5 Track A: validation on known generators

Track A is the core executable map of Doc. 2's strata. These notebooks use known finite-state generators and exact numerical linear algebra. Their primary purpose is to make the signature tuple concrete.

5.1 Notebook 01: rank-one scalar compression

Purpose. This notebook instantiates the monograph's scalar regime. In Doc. 1 Sections V.2 and Appendix D, a block becomes an admissible scalar input to the outer evaluator when the principal

killed mode dominates on the cycle timescale. Operationally, if $g = a_2 - a_1$ is the compression gap, scalar compression requires $g\Delta \gg 1$. Doc. 2 identifies this as stratum 1.

Model. Use a finite reversible killed chain with a simple principal eigenvalue and a visible spectral gap. The simplest version can use the Gibbs-lifted one-well birth-death block from Doc. 1 Part IV and Appendices C–D. A lighter initial version can use a hand-built tridiagonal killed generator with a clear principal mode.

Diagnostics. Compute decay rates $a_1, a_2, \dots$, the gap product $(a_2 - a_1)\Delta$, the survival profile $q_\Delta = e^{\Delta G} 1$, the rank-one reconstruction, and the scalarity defect $\varepsilon_{\text{scalar}}$ under declared readouts. Vary Δ and the potential/barrier parameters.

Why it is needed. The whole program depends on the claim that scalar interface values are not merely assigned labels. They arise when the killed semigroup has actually compressed. This notebook makes scalar interface formation visible and gives a numerical standard for later failures.

Hoped-for output. A phase diagram showing a scalarity region where $g\Delta$ is large and $\varepsilon_{\text{scalar}} \leq \eta$, plus boundary plots showing how scalarity degrades as $g\Delta$ approaches order one.

5.2 Notebook 02: one-well formal-cone non-diagnostic case

Purpose. This notebook targets the subtle region between stratum 1 and interpretable stratum 2a. Doc. 2 distinguishes formal positive-cone structure from interpretable basin-witnessed structure. A one-well killed chain at intermediate Δ should retain additional modes, but those modes are intrawell relaxation modes rather than basin indicators.

Model. Use a one-well Gibbs or diffusive killed chain. Choose Δ so that the second mode remains visible: $(a_2 - a_1)\Delta = O(1)$ while the tail modes are suppressed. The retained band is low-dimensional, but there is no multi-basin geometry.

Diagnostics. Compute the retained modes, sign structure, scalarity defect, band defect, eventual or formal cones induced by the retained band, and the failure of basin-witness criteria. Track how the second mode changes with threshold width, barrier shape, and cycle length.

Why it is needed. The formal-cone non-diagnostic region is doing important conceptual work in Doc. 2. It says that mathematical positivity alone is too cheap to carry biological interpretation. This notebook should make that distinction operational: a cone-like ordering may exist, but no basin witness is present.

Hoped-for output. A controlled example where $K > 1$, $\varepsilon_{\text{band}} \leq \eta$, $\varepsilon_{\text{scalar}} > \eta$, formal cone structure can be constructed, and the interpretability witness fails. This gives a computational anchor for the non-diagnostic region.

5.3 Notebook 03: multi-well interpretable stratum 2a

Purpose. This notebook builds the positive case for interpretable stratum 2a. Doc. 2 Section 5 ties interpretable 2a to metastable-basin geometry: retained modes should recombine into basin-local survival profiles or basin indicators, and a basin-mass readout should be canonical relative to the viability question.

Model. Use reversible finite-state chains with two or three metastable wells and killing at viability-failure boundaries. The cleanest first implementation is a one-dimensional or grid-based potential landscape with multiple wells and low-temperature transitions.

Diagnostics. Compute the low spectrum, identify metastable basins, recombine eigenvectors into basin-local profiles, construct the basin cone, compute basin-mass readouts, and test cycle-window stability. Compare direct spectral modes with recombined basin coordinates.

Why it is needed. Interpretable 2a must not remain a citation to metastability theory. It must become a computable signature regime. This notebook demonstrates that finite-band structure can have biological or dynamical meaning when the retained coordinates correspond to basin-local persistence.

Hoped-for output. A family of examples where basin labels are stable, the retained band has $K \geq 2$, scalar compression fails, and a canonical basin-weighted scalarization is supplied by the model geometry. This validates the narrow-but-nonempty status of interpretable 2a.

5.4 Notebook 04: generic cone-valued stratum 3

Purpose. Doc. 2 upgrades generic finite-band non-scalar systems from “failure” to stratum 3: vector/cone-valued comparison without a canonical scalar. This notebook makes forced pluralism visible.

Model. Use small finite-dimensional retained-band examples with an explicitly declared cone K and admissible readout family $\mathcal{R} \subseteq K^*$. The system can be abstract at first: finite vectors representing retained interface states, optionally generated by a killed semigroup with signed higher modes.

Diagnostics. Sample pairs x, y of retained interface states. Check whether $x - y \in K$, $y - x \in K$, or neither. Compare rankings under the full dual cone K^* and under declared subfamilies $\mathcal{R}$. Identify pairs ranked unanimously by $\mathcal{R}$ but not by the full cone, and pairs whose ranking depends on scalarization.

Why it is needed. The forced-pluralism thesis is mathematically precise but easy to misread as indecision. This notebook shows the substantive content: cone-comparable pairs are ordered; cone-incomparable pairs require a declared scalarization priority.

Hoped-for output. Cone-comparability diagrams and ranking maps showing where the dynamics themselves order pairs and where the declared readout class supplies additional priorities.

5.5 Notebook 05: readout rigidity and thin-domain escape

Purpose. Doc. 2 Section 4 dissolves the proposed readout-insensitive finite-band middle. Theorem 4.1 says that ranking-preserving linear readouts on rich domains are projectively equivalent. The point is simple linear algebra, but readers benefit from seeing the geometry.

Model. Construct rich comparison domains with nonempty relative interior and thin domains such as curves, rays, low-dimensional faces, or empirical clouds. Generate linear readouts and numerically test strict-order agreement.

Diagnostics. For rich domains, impose or search for ranking agreement and observe projective collapse. For thin domains, display apparent nontrivial agreement and show that it disappears when the domain is thickened.

Why it is needed. This notebook prevents a common misinterpretation of the dissolution result. Thin empirical domains can mimic a 2b-like middle. The theorem says the middle does not survive on rich comparison domains. Computation can show both facts at once.

Hoped-for output. A geometric demonstration that finite-band readout-insensitivity is not robust; it either collapses to a single projective readout or relies on a domain too thin to support the claimed stratum.

5.6 Notebook 06: dynamic J -axis regimes

Purpose. Doc. 2's dynamic axis $J \in \{qs, fz, int\}$ refines the static stratum rather than replacing it. Doc. 1 Section V.3 and Appendix F realize dynamic scalar interface values as Feynman–Kac quantities on a slow coarse state. This notebook shows how quasi-static, fast-coarse-mixing, and intermediate regimes differ computationally.

Model. Use a finite coarse state $z \in Z$, with a z -dependent killed generator G_z or a local principal rate $a_1(z)$ after compression. Vary the coarse transition rate relative to the compression gap, killing scale, and cycle time.

Diagnostics. Compute quasi-static approximations, fast- z averaged approximations, and the full intermediate Feynman–Kac value

$$A(\Delta, z) = \left[e^{\Delta(Q - \text{diag}(a_1))} \mathbf{1} \right]_z.$$

Measure Jensen gaps, dynamic scalarity defects, mode-crossing sensitivity, and basin-label instability.

Why it is needed. The dynamic axis is a genuine part of the signature type. Without computational examples, J risks reading as a label rather than a continuation problem. This notebook clarifies what changes when the static retained structure is transported through time.

Hoped-for output. Regime diagrams showing where quasi-static and fast-mixing reductions work, where intermediate Feynman–Kac effects dominate, and where dynamic motion destroys scalar or basin-witness continuity.

6 Track B: perturbative boundary diagnostics

Track B is motivated by Doc. 1 Appendix H and Memos 20–21. It moves the program beyond exact non-transferability. Real boundaries are porous. The question is whether porosity matters in the spectral and survival-weighted coordinates relevant to AM–GM aggregation.

6.1 Notebook 07: weak-coupling scalar compression integrity

Purpose. Doc. 1 Appendix H.2 and Memo 20 define the first weak-coupling channel: scalar compression integrity. A block with unperturbed generator G_0 and simple principal mode is perturbed to $G_t = G_0 + E_t$. Scalarity survives when leakage from the principal mode into the complement is small relative to the surviving compression gap.

Model. Start with the minimal two-mode leaky block of Appendix H.3. Then move to finite Gibbs blocks. Let P_1 be the principal projection, $Q_1 = I - P_1$, and define the leakage rate

$$\lambda_{\perp}(t) = \|Q_1 E_t P_1\|.$$

Vary leakage strength, gap drift, cycle length, and moving-landscape forcing.

Diagnostics. Compute $\lambda_{\perp}$, surviving gap $\tilde{g}$, moving-frame forcing q , and scalarity defect. Test the criterion

$$\frac{\lambda_{\perp} + q}{\tilde{g}} \ll 1.$$

Plot the transition from scalar stratum 1 to finite-band behavior as the leakage-to-gap ratio grows.

Why it is needed. This notebook turns exact boundary separation into perturbative boundary integrity. It answers a biologically important question: not whether coupling exists, but whether coupling contaminates the principal interface mode before the outer evaluator acts.

Hoped-for output. A numerical phase diagram with axes such as $g\Delta$, $\lambda_{\perp}/\tilde{g}$, and $\varepsilon_{\text{scalar}}$, plus validation of the two-mode analytical bound.

6.2 Notebook 08: Feynman–Kac product defects under coupling

Purpose. Doc. 1 Appendix H.4 and Memo 21 define the second weak-coupling channel: composition integrity. Suppose scalar interface values already exist on block coarse spaces. Weak coupling may still prevent the raw product Feynman–Kac object from factoring into marginal values.

Model. Use a finite product coarse process with uncoupled generator H_0 and weakly coupled generator

$$H_{\varepsilon} = H_0 + \varepsilon B.$$

For nonnegative readouts ℓ, r , compute

$$A_{\varepsilon}(\Delta) = \ell^{\top} e^{\Delta H_{\varepsilon}} r.$$

Compare this to the uncoupled product of marginal Feynman–Kac values.

Diagnostics. Compute the raw product-correlation defect

$$\varepsilon_{\text{prod}}(\varepsilon) = |\log A_\varepsilon(\Delta) - \log A_0(\Delta)|.$$

Compute the tilted sensitivity density

$$\frac{\ell^\top e^{(\Delta-s)H_\varepsilon} B e^{sH_\varepsilon} r}{\ell^\top e^{\Delta H_\varepsilon} r},$$

which measures coupling under the survival-biased Feynman–Kac bridge.

Why it is needed. Raw coupling norm is not the right diagnostic. Coupling where surviving paths concentrate matters more than coupling in high-killing regions that almost no survivor visits. This notebook makes the survival-biased nature of composition defects explicit.

Hoped-for output. Examples where two couplings with the same operator norm have different AM–GM significance because the Feynman–Kac tilted bridge weights them differently.

6.3 Notebook 09: FEP and Markov-blanket cross-diagnostics

Purpose. Doc. 1 Section I.4 and Appendix H.9 sharpen the relation to the Free Energy Principle and Markov blankets. A blanket is not AM–GM-significant merely because it is a conditional-independence boundary. It matters for viability aggregation when its porosity is small relative to the spectral compression gap and small enough under the survival-biased Feynman–Kac bridge not to destroy product composition.

Model. Use canonical Active Inference or Markov-blanket toy models with explicit internal, blanket, and external states. Add absorbing failure or viability readouts. Construct candidate partitions suggested by the blanket structure and compute the corresponding AM–GM diagnostics.

Diagnostics. Measure raw blanket porosity, projected leakage $\lambda_\perp$, compression gap g or $\tilde{g}$, scalarity defect, Feynman–Kac product defect, and implied stratum. Compare boundary assessments produced by blanket structure and AM–GM signature type.

Why it is needed. The program claims to provide a type discipline for individuality-as-persistence, not to replace FEP on its own terms. A cross-diagnostic notebook makes that relationship concrete. It can show where a Markov blanket supports scalar interface formation, where it supports only finite-band or cone-valued signatures, and where it is irrelevant to viability aggregation.

Hoped-for output. A set of disagreement cases: blanket-present but no scalar interface; weak or ambiguous blanket but robust scalar or interpretable 2a signature; blanket porosity small in raw terms but large in spectral/Feynman–Kac coordinates; raw coupling large but survival-tilted defect small.

7 Track C: exploratory cross-stratum composition

7.1 Notebook 10: cross-stratum composition strategies

Purpose. Doc. 2 Section 9.1 lists F4 as an open arc. Same-type composition can be constructed under strong shared-coordinate hypotheses, as in Doc. 2 Example 7.2. Cross-stratum composition is more delicate: one may need to scalarize, embed, retain a product type, or prove promotion to stratum 3. This notebook explores those strategies before a theorem is written.

Model. Build composite systems mixing stratum-1 scalar blocks, interpretable-2a basin-vector blocks, and stratum-3 cone-valued blocks. Begin with compatible same-type 2a coordinate pooling, then deliberately break compatibility: mismatched basin labels, different cone choices, scalar blocks embedded into vector spaces, and product-typed pairs (c, v) .

Diagnostics. For each conversion strategy, measure information loss, cone-order preservation, scalar ranking stability, regrouping behavior, and sensitivity to the supplied compatibility data C . Compare direct composition to two-step composition under cluster regrouping.

Why it is needed. F4 is unlikely to be solved by pure abstraction first. The space of type-conversion choices is too large and too compatibility-dependent. Computational examples can reveal which strategies are stable and which smuggle in hidden scalarization.

Hoped-for output. A taxonomy of composition failures and admissible conversions: when scalarization is harmless, when embedding is arbitrary, when product-typed signatures are unavoidable, and whether generic promotion to stratum 3 is theorem-shaped or overambitious.

8 Track D: estimation and Document 3 prototypes

Track D turns known-generator diagnostics into finite-sample methodology. It is the runway to Document 3.

8.1 Notebook 11: signature estimation from trajectories

Purpose. Doc. 2 repeatedly states that its typology classifies supplied signatures rather than extracting them. This notebook starts the extraction problem under controlled conditions.

Model. Use the same systems as Track A, but sample trajectories instead of exposing the generator. Allow full observation first, then partial observation and noise.

Diagnostics. Estimate generators or semigroups, low eigenvalues, retained dimension K , scalarity defect, band defect, and dynamic regime label. Compare estimated signatures $\hat{\Sigma}$ against true signatures Σ .

Why it is needed. Without estimation, the program cannot move from formal typology to data. This notebook establishes the benchmark loop: known truth, finite samples, recovered signature, uncertainty.

Hoped-for output. Finite-sample error curves for signature components, confidence intervals for stratum assignment, and failure cases where scalarity or basin witnesses are falsely inferred.

8.2 Notebook 12: basin reconstruction and hysteresis diagnostics

Purpose. Interpretable 2a depends on basin-witness data. Doc. 2 Section 9.2 names basin reconstruction and hysteresis/multistability assays as central empirical methods. This notebook tests whether those data can be recovered from trajectories and perturbations.

Model. Use multi-well chains with known basins, metastable switching, and controlled perturbations. Vary barrier height, sampling frequency, noise, and observation coarseness.

Diagnostics. Recover basins using transition statistics, committor-like quantities, clustering, or perturbation response. Test basin-label stability over cycle windows. Detect hysteresis by sweeping control parameters forward and backward.

Why it is needed. The hardest empirical distinction in Doc. 2 is formal-cone non-diagnostic versus interpretable 2a. That distinction becomes meaningful only if basin witness data can be estimated.

Hoped-for output. A benchmark for when basin witnesses are identifiable and when finite-band structure should remain non-diagnostic or stratum 3.

8.3 Notebook 13: cone-comparability tests from samples

Purpose. Doc. 2 Section 6 gives stratum 3 its mathematical content through cone order and admissible scalarizations. Empirically, this requires testing whether observed comparisons are unanimous over $\mathcal{R}$ or scalarization-dependent.

Model. Use synthetic retained-band systems with known cone K and readout class $\mathcal{R}$, then generate noisy samples of interface states and comparisons.

Diagnostics. Estimate cone membership, dual-cone inequalities, unanimous- $\mathcal{R}$ comparison, and scalarization-dependent disagreements. Measure false cone-comparable and false cone-incomparable rates.

Why it is needed. Forced pluralism is not empirically useful unless one can detect the difference between structural partial order and measurement-imposed scalar ranking.

Hoped-for output. Operational tests for cone-comparability and scalarization dependence that can become part of Document 3.

9 Implementation architecture

9.1 The signature object

The computational leg should use a common signature record matching Doc. 2 Section 2:

$$\Sigma = (K, a, \varepsilon_{\text{scalar}}, \varepsilon_{\text{band}}, \mathcal{R}, C, \eta; J).$$

The record should also store provenance metadata: model class, generator family, readout, partition, cycle length, tolerance, basis convention, cone convention, numerical precision, and whether the signature was evaluated from a known model or estimated from data.

9.2 Core diagnostic modules

The following modules are conceptual rather than code-level requirements:

1. **Spectral module.** Compute eigenvalues, spectral gaps, retained projections, Riesz-projection continuity diagnostics, and mode-crossing warnings.
2. **Defect module.** Compute scalarity defect, band defect, dynamic defect, weak-coupling leakage defect, and product-correlation defect, with normalization conventions attached.
3. **Cone module.** Construct or accept cones, dual cones, admissible readout families, and pairwise comparability tests.
4. **Basin module.** Detect or verify metastable basins, basin-local profiles, basin-mass readouts, and cycle-window stability.
5. **Composition module.** Evaluate scalar geometric pooling, coordinatewise same-type pooling, product-typed retention, scalarization-before-composition, and embedding strategies.
6. **Estimation module.** Estimate signatures from trajectories, perturbations, and partial observations, with uncertainty.
7. **Assignment module.** Apply Doc. 2’s assignment schema and precedence convention, recording which conditions triggered the assignment.

9.3 Assignment reports

Every notebook should produce an assignment report with four layers:

1. raw model or data description;
2. computed or estimated signature;
3. stratum assignment with active inequalities and witness data;
4. disagreement log and sensitivity to η , $\mathcal{R}$, and C .

This report is more important than any single plot. It preserves the type discipline and makes the computational result auditable.

10 Milestones

10.1 Phase 1: known-generator validation

Build Notebooks 01–05. The deliverable is a stable evaluation pipeline and a small gallery of canonical strata. Success criterion: each of Doc. 2’s static regions has at least one controlled model whose signature assignment is reproducible and whose boundary behavior can be varied.

10.2 Phase 2: dynamic and weak-coupling diagnostics

Build Notebooks 06–09. The deliverable is a dynamic and perturbative extension of the signature lab. Success criterion: J -axis behavior, leakage-to-gap scalarity loss, Feynman–Kac product defects, and FEP/blanket cross-diagnostics are all computable in small finite-state systems.

10.3 Phase 3: cross-stratum composition

Build Notebook 10. The deliverable is an exploratory map of F4. Success criterion: the lab identifies which conversion strategies preserve structural information, which lose partial-order content, and which require hidden compatibility assumptions.

10.4 Phase 4: estimation prototypes

Build Notebooks 11–13. The deliverable is the first Document 3 prototype. Success criterion: finite-sample trajectories can recover at least coarse stratum assignment in the canonical systems, with uncertainty and failure modes made explicit.

10.5 Phase 5: theory feedback

Write a short computational findings note after the first full pass. Its purpose is not to publish a software paper immediately. Its purpose is to feed results back into Doc. 2 and the F1–F4 arcs: sharpen hypotheses, identify edge cases, and decide which computational phenomena deserve theorem treatment.

11 How the computational leg informs the mathematics

11.1 Feedback to Doc. 2

The computational leg should test whether Doc. 2’s regions are operationally separable. The most important separations are:

- stratum 1 versus finite-band non-scalar behavior as $g\Delta$ decreases;
- formal-cone non-diagnostic versus interpretable 2a;
- interpretable 2a versus stratum 3 under dynamic or weak-coupling deformation;
- cone-comparable versus scalarization-dependent comparison in stratum 3;
- same-type composition versus cross-type composition requiring C -data.

If these distinctions are numerically brittle even in synthetic systems, the paper should say so. If they are robust under clear margins, the computational leg can suggest margin hypotheses for future theorems.

11.2 Feedback to Doc. 1 Appendix H

Appendix H turns weak coupling into two projected conditions: scalar leakage relative to the compression gap, and Feynman–Kac tilted coupling sensitivity relative to product composition. The computational leg can test how these quantities behave in finite models and whether they predict stratum transitions. This is where the program starts to replace binary non-transferability with perturbative boundary integrity.

11.3 Feedback to F1–F4

F1: weak-coupling cone and basin stability. The lab should test whether cones, dual cones, basin labels, and basin-mass readouts survive weak coupling when the retained spectral subspace itself remains stable. This will indicate whether polyhedral angular margins and basin return-time bounds are the right hypotheses.

F2: matrix-valued Feynman–Kac. The dynamic J -axis notebook should produce small examples where scalar Feynman–Kac continuation fails and retained-band transport is needed. These examples can guide the minimal assumptions for a matrix-valued Feynman–Kac theory: fixed rank, non-crossing, gauge choices, and path-ordered continuation.

F3: dynamic stability of interpretable 2a. The lab should search for both positive and negative examples: cases where basin labels persist under intermediate dynamics, and cases where retained subspaces persist but basin canonicity is lost. This is likely the most valuable computational input to F3.

F4: cross-stratum composition. The lab should map which type conversions preserve comparison content. This may reveal that a generic F4 theorem is too ambitious, while a disciplined conversion calculus is achievable.

12 How the computational leg moves toward empirics

The computational leg should make Document 3 less abstract in three ways.

First, it supplies benchmark systems with known signatures. Any empirical method should be tested on these before being applied to biological systems. Second, it supplies uncertainty-aware estimators for signature components. A biological claim should eventually report confidence in K , $\varepsilon_{\text{scalar}}$, $\varepsilon_{\text{band}}$, basin witnesses, cone comparisons, and J , not just a point stratum. Third, it supplies disagreement diagnostics. If a biological system is assigned differently under different readout classes $\mathcal{R}$ or tolerances η , that is not necessarily a failure. It may be the empirical face of the typology’s index dependence.

The long-term empirical pathway is therefore:

synthetic generator $\rightarrow$ computed signature $\rightarrow$ sampled trajectories
 $\rightarrow$ estimated signature $\rightarrow$ biological data $\rightarrow$ indexed signature profile.

The last object is not a single answer to “is this an individual?” It is a map from partition, timescale, and viability readout to signature type.

13 Risks and guardrails

13.1 Risk 1: demo drift

Pretty plots can become a substitute for theory. Guardrail: every notebook must state the formal claim it validates, the open question it explores, and what would count as pressure on the theory.

13.2 Risk 2: hidden scalarization

It is easy to impose a scalar readout on a finite-band object and then treat the result as canonical. Guardrail: every scalarization must be recorded in $\mathcal{R}$ or C , and plots should distinguish canonical scalar values from supplied scalar readouts.

13.3 Risk 3: overclaiming FEP comparisons

A Markov blanket and an AM–GM interface signature answer different questions. Guardrail: the FEP crosswalk should compare diagnostics, not declare winners. The correct claim is that AM–GM diagnoses which candidate blankets support viability aggregation of a given type.

13.4 Risk 4: estimation optimism

Finite-sample spectral estimation can be fragile. Guardrail: the estimator pipeline must report uncertainty, sensitivity to observation models, and failure rates, especially near stratum boundaries.

13.5 Risk 5: coupling mistaken for failure

Nonzero coupling does not automatically destroy a boundary. Guardrail: weak coupling should be evaluated through projected leakage and survival-weighted product defects, not raw coupling alone.

14 Recommended first deliverable

The first deliverable should be a repository and a short internal findings memo, not a public paper. The repository should contain Notebooks 01–05 and the common signature-report machinery. The findings memo should answer:

1. Which signature components were straightforward to compute?
2. Which required arbitrary conventions?
3. Which stratum distinctions were robust under parameter variation?
4. Which distinctions were sensitive to η , $\mathcal{R}$, or C ?
5. Which computational edge cases should become formal lemmas or counterexamples?

After that, the weak-coupling and FEP notebooks should be prioritized because they have the highest external leverage. The estimation notebooks should begin once the evaluator pipeline is stable enough to provide ground truth.

15 Closing

The computational leg is needed because the program’s mature object is now a signature, not a slogan. Document 1 shows how scalar interface values can be realized in a killed-process class. Document 2 shows that scalarity is only one region of a finite-band typology. Appendix H of Document 1 turns exact non-transferability into perturbative boundary integrity. The future Document 3 must estimate signatures from data.

A computational signature diagnostics suite is the missing bridge. It will make the typology executable, reveal which formal hypotheses are sharp, test weak-coupling and dynamic failure modes, explore cross-stratum composition before theory outruns examples, and create the benchmark loop

needed for empirical extraction. Its purpose is not to decorate the program with simulations. Its purpose is to make the program falsifiable, portable, and empirically preparable.

A Notebook map

Notebook	Primary target	Main question
01	Rank-one scalar compression	When does a known killed generator produce a positive scalar interface value?
02	One-well formal cone	How does finite-band structure appear without basin interpretation?
03	Multi-well interpretable 2a	When do low modes recombine into basin-local signatures and canonical basin readouts?
04	Stratum 3 cone order	Which pairs are structurally comparable, and which require scalarization?
05	Readout rigidity	Why does a readout-insensitive finite-band middle collapse on rich domains?
06	Dynamic J axis	How do quasi-static, fast- z , and intermediate Feynman–Kac regimes change the signature?
07	Weak-coupling scalar integrity	When does leakage destroy scalar compression?
08	Feynman–Kac product defects	When do already-compressed scalar blocks fail to factor under weak coupling?
09	FEP/blanket crosswalk	Which candidate blankets support AM–GM interface signatures?
10	Cross-stratum composition	What do scalarize, embed, product-type retention, and promotion strategies cost?
11	Signature estimation	Can finite trajectories recover known-generator signatures?
12	Basin reconstruction	Can interpretable 2a witness data be recovered from samples and perturbations?
13	Cone tests from samples	Can cone-comparability and scalarization-dependence be distinguished empirically?

B Minimal repository structure

A minimal repository should separate theory-facing notebooks, estimation notebooks, shared diagnostics, and reports:

- `notebooks/validation/`: Notebooks 01–06.
- `notebooks/perturbative/`: Notebooks 07–09.

- `notebooks/composition/`: Notebook 10.
- `notebooks/estimation/`: Notebooks 11–13.
- `signature/`: shared signature objects, defect diagnostics, assignment reports.
- `models/`: finite killed chains, Gibbs wells, multi-well systems, cone examples, weak-coupling products.
- `reports/`: rendered assignment reports and disagreement logs.

C Internal references

1. **Document 1, monograph v34.** Relevant sections: §I.4 for neighboring programs and the FEP/blanket relation; §I.5 for program placement and formal-status convention; Part IV and Appendices B–C for diffusive baseline and Gibbs lift; Part V and Appendices D–G for scalar synthesis, dynamic Feynman–Kac values, and regrouping; Appendix H for weakly coupled boundaries and perturbative boundary integrity; Appendix I for near-degenerate finite-mode interface structure.
2. **Document 2, finite-band typology v16.** Relevant sections: §1 for the rank-one-to-signature reframe; §2 for the signature object and assignment schema; §3 for static strata; §4 for readout rigidity and A5-admissible classification; §5 for interpretable 2a; §6 for stratum 3 cone order; §7 for operational defects and same-type composition; §8 for the dynamic axis; §9 for F1–F4 and Document 3 implications.
3. **Memo 20.** Scalar Compression Under Weak Coupling. Supplies the leakage-to-gap channel and the criterion $(\lambda_{\perp} + q)/\tilde{g} \ll 1$.
4. **Memo 21.** Feynman–Kac Product Defects Under Weak Coupling. Supplies the tilted sensitivity identity and the raw product-correlation defect.